

NHS Patient Records Data Quality Analysis

Portfolio Project Report

Tools: Python • Pandas • MySQL • SQL • Power BI • Excel/CSV

Dataset: 10,000 synthetic NHS patient records

1. Executive Summary

This project examines the quality of a synthetic NHS patient dataset containing 10,000 patient records.

The purpose of the project was to identify data-quality problems before the data was used for reporting and further analysis. I used **Python and Pandas** to profile and validate the dataset, **MySQL and SQL** to analyse the cleaned data, and **Power BI** to present the findings through an interactive dashboard.

The analysis produced an overall **Data Quality Score of 97.01%**.

Although the dataset performed strongly in terms of validity and uniqueness, the analysis identified several areas requiring attention, particularly missing information and inconsistent admission and discharge dates.

2. Project Objectives

The project aimed to:

- Assess data completeness
- Validate NHS Numbers
- Validate Patient IDs
- Check admission and discharge date consistency
- Identify duplicate records
- Validate categorical fields
- Identify missing values
- Calculate an overall Data Quality Score
- Clean the dataset
- Analyse the cleaned data using SQL
- Communicate findings through Power BI

3. Dataset

The dataset contains **10,000 synthetic patient records** representing a healthcare-style patient information system.

The fields analysed include:

- Patient ID
- NHS Number
- Gender
- Date of Birth
- Admission Date
- Discharge Date
- GP Practice
- Postcode
- Diagnosis Code
- Department
- Appointment Status

The dataset was created for educational and portfolio purposes and does not contain real patient information.

4. Technology Stack

Technology Role in the project

Python	Data-quality analysis and cleaning
Pandas	Data manipulation and validation
Excel/CSV	Dataset storage and preparation
MySQL	Database storage
SQL	Further analysis and investigation
Power BI	Dashboard and visualisation
DAX	Power BI calculations
GitHub	Portfolio and project documentation

5. Data Quality Methodology

The project assessed four main dimensions of data quality:

Completeness

Checks whether important fields contain values.

Validity

Checks whether values follow the expected format or rules.

Uniqueness

Checks whether identifiers are duplicated where they should be unique.

Consistency

Checks whether related fields make sense together.

The results from these checks were combined to produce an overall Data Quality Score.

6. Data Quality Results

Data Quality Dimension	Score
Completeness	93.8%
Validity	99.1%
Uniqueness	99.9%
Consistency	95.0%
Overall Data Quality	97.0%

The results show that the dataset was generally of good quality.

However, the overall score should not be interpreted as meaning that every record is error-free. The detailed checks identified specific records and fields that require attention.

7. Missing-Value Analysis

The missing-value analysis identified several areas of concern.

Field	Missing Records	Missing %
Gender	2,065	20.65%
GP Practice	1,722	17.22%
Diagnosis Code	1,292	12.92%
Postcode	1,291	12.91%
NHS Number	397	3.97%

Finding

Gender was the largest missing-value issue, with 2,065 records affected.

GP Practice was the next largest issue, with 1,722 missing values.

These gaps could affect demographic analysis, GP-level analysis and some clinical reporting.

8. NHS Number Validation

NHS Numbers were checked for format, missing values and duplicates.

The format validation achieved **100% validity**, meaning no incorrectly formatted NHS Numbers were identified.

However:

- **397 NHS Numbers were missing**
- **9 duplicate NHS Number occurrences were identified**

The duplicate NHS Number records should be investigated to establish whether they are caused by data-entry errors or identifier problems.

9. Patient ID Validation

Patient IDs performed strongly during validation.

The analysis found:

- No missing Patient IDs
- No invalid Patient IDs
- No duplicate Patient IDs

This means Patient ID provided a reliable unique identifier for the records in the dataset.

10. Date Consistency

Admission and discharge dates were checked to ensure that they followed a logical sequence.

The analysis identified:

491 inconsistent records

In these records, the discharge date occurred before the admission date.

This could affect calculations such as:

- Length of stay
- Patient-flow analysis
- Admission trends
- Discharge trends
- Time-based KPIs

These records were therefore flagged for further investigation.

11. Categorical Validation

Categorical fields were checked to ensure that unexpected values were not present.

The analysis included:

- Gender
- Department
- Appointment Status

The relevant validity checks achieved **100%**, following standardisation.

It is important to distinguish between **validity and completeness**. For example, Gender values could be valid where present while still having a significant number of missing values.

12. Python and Pandas Analysis

Python was used as the main data-quality analysis tool.

Pandas was used to:

- Load the dataset
- Profile the data
- Identify missing values
- Validate fields
- Check duplicate records
- Analyse NHS Numbers
- Validate Patient IDs
- Check date relationships
- Calculate data-quality measures
- Create validation flags
- Produce the cleaned dataset

The cleaning process focused on **flagging issues rather than simply deleting records**.

This approach makes the analysis more transparent and allows problematic records to be investigated later.

13. Data Cleaning

The cleaning workflow was:

Raw Dataset

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Load into Python/Pandas

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Data Profiling

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Validation Checks

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Identify Data-Quality Issues

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Create Quality Flags

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Clean Dataset

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Export to CSV

The cleaned dataset was then prepared for database analysis.

14. SQL Analysis

The cleaned dataset was imported into **MySQL**.

SQL was then used to investigate the data from a database perspective.

The analysis included:

- Total patient records
- Missing values
- Valid and invalid records
- Duplicate NHS Numbers
- Data-quality issues
- Department-level analysis
- Quality measures

This stage demonstrated the ability to move from Python-based data preparation into a relational database environment.

15. Power BI Dashboard

Power BI was used to communicate the findings to a wider audience.

The dashboard includes key metrics such as:

- Overall Data Quality Score
- Valid Records
- Invalid Records
- Missing Values
- Data Quality Issues

Additional visuals were developed to analyse:

- Data quality by field
- Gender quality
- Patient ID quality
- Date quality
- Data-quality issues
- Department-level results

Slicers were also included to allow users to explore the results interactively.

16. Key Insights

The analysis identified several important findings.

Overall quality was strong

The dataset achieved an overall score of **97.01%**.

Completeness was the weakest area

Completeness scored **93.85%**, mainly because of missing Gender, GP Practice, Diagnosis Code and Postcode values.

Gender requires the most attention

There were **2,065 missing Gender values**, representing 20.65% of the dataset.

Date consistency needs improvement

There were **491 records** where the discharge date occurred before the admission date.

Patient IDs were reliable

Patient IDs were complete and unique across the dataset.

NHS Number issues remain

Although NHS Number format validation was successful, missing and duplicate NHS Numbers require further investigation.

17. Recommendations

Based on the analysis, I recommend the following:

1. Investigate the 491 inconsistent admission and discharge dates.
2. Review records with missing NHS Numbers.
3. Investigate duplicate NHS Number occurrences.
4. Improve the collection of Gender information.
5. Improve the completeness of GP Practice, Diagnosis Code and Postcode fields.
6. Introduce validation rules when data is entered or imported.
7. Retain data-quality flags so issues remain traceable.
8. Repeat the data-quality checks whenever the dataset is refreshed.
9. Only use appropriately validated data for downstream reporting.
10. Monitor data quality regularly rather than treating it as a one-off exercise.

18. Project Workflow

The complete project follows this workflow:

Raw Patient Dataset



Python & Pandas



Data Quality Assessment



Data Cleaning



Cleaned Dataset



MySQL



SQL Analysis



Power BI

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Dashboard

↓

Insights & Recommendations

19. Project Structure

NHS-Data-Quality-Analysis/

```
|
|
|─ data/
|   |─ NHS_patient_records.xlsx
|   |─ NHS_patient_records_cleaned.csv
|   └─ data_dictionary.xlsx
|
|─ python/
|   |─ data_quality_analysis.py
|   └─ data_cleaning.py
|
|─ sql/
|   |─ NHS_data_quality_analysis.sql
|   └─ README.md
|
|─ powerbi/
|   └─ NHS_Data_Quality_Dashboard.pbix
|
|─ reports/
|   └─ NHS_Data_Quality_Analysis_Report.pdf
|
└─ README.md
```

20. Skills Demonstrated

This project demonstrates practical experience in:

- Python
- Pandas
- Data cleaning
- Data validation
- Data profiling
- Missing-value analysis
- Duplicate detection
- Data-quality measurement
- MySQL
- SQL
- Power BI
- DAX
- Data visualisation
- Dashboard development
- Data storytelling
- GitHub documentation

21. Portfolio Value

This project demonstrates an **end-to-end data analyst workflow** rather than simply producing a dashboard.

It shows how I:

Identify a data problem → investigate the data → validate and clean it → analyse it with SQL → build a Power BI dashboard → communicate insights and recommendations.

22. Conclusion

The NHS Data Quality Analysis demonstrates that the synthetic dataset is generally of good quality, achieving an overall score of **97.01%**.

However, the analysis identified clear areas for improvement, particularly missing information and inconsistent admission and discharge dates.

Using Python and Pandas for validation, MySQL and SQL for further analysis, and Power BI for reporting provided a complete analytical workflow.

The project has therefore provided practical experience in **data quality, data cleaning, database analysis, visualisation and communicating findings to non-technical users.**

Disclaimer

This project uses **synthetic/anonymised data created solely for educational and portfolio purposes.**

It does not contain real NHS patient information or real patient records.